PROPERTIES OF REPRESENTABLE MORPHISMS ARE UNCHANGED BY THE CHANGE OF BASE EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one identifying equality; it is part of the counted proof span and is preserved verbatim.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- The change of base lemma for localization functors. Lemma lemma-change-base-scheme (2949-2979, counted as L44175), used at proof lines 3160-3162. Makes U' -> F' surjective and etale.
- The equivalence between spaces over S and pairs over S'. Lemma lemma-category-of-spaces-over-smaller-base-scheme (3065-3087, counted as L44179), used at proof lines 3150-3152. Produces f' from f.
- Criterion for properties of representable morphisms of spaces. Lemma lemma-representable-morphisms-spaces-property, used at proof lines 3170-3171. Reduces P(f) to the induced morphism of schemes.
- Definition of the relative representable property type. Definition definition-relative-representable-property, used at proof lines 3153-3155. Fixes the class of properties P to which the statement applies.

Source reference. Original definition 025V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2828–2828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04SG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6219–6219. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03I4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4590–4590. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3822–3822. Navigation pointer into the Stacks

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let Sch_{fppf} be a big fppf site. Let $S \to S'$ be a morphism of this site. Let F be an algebraic space over S. Let T be a scheme over S and let $f : T \to F$ be a morphism over S. Let $f' : T' \to F'$ be the morphism over S' we get from f by applying the equivalence of categories described in Lemma 04SG. For any property $\mathcal{P}$ as in Definition 025V we have $\mathcal{P}(f') \Leftrightarrow \mathcal{P}(f)$.*

Proof. Suppose that U is a scheme over S, and $U \to F$ is a surjective étale morphism. Denote U' the scheme U viewed as a scheme over S'. In Lemma 03I4 we have seen that $U' \to F'$ is surjective étale. Since

$$j(T \times_{f,F} U) = T' \times_{f',F'} U'$$

the morphism of schemes $T \times_{f,F} U \to U$ is identified with the morphism of schemes $T' \times_{f',F'} U' \to U'$. It is the same morphism, just viewed over different base schemes. Hence the lemma follows from Lemma 02WZ. □